

GDB's Obsolete Annotations

Edition 1.0
July 2003

Free Software Foundation

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1 What is an Annotation?

To produce obsolete level two annotations, start GDB with the `--annotate=2` option.

Annotations start with a newline character, two ‘`control-z`’ characters, and the name of the annotation. If there is no additional information associated with this annotation, the name of the annotation is followed immediately by a newline. If there is additional information, the name of the annotation is followed by a space, the additional information, and a newline. The additional information cannot contain newline characters.

Any output not beginning with a newline and two ‘`control-z`’ characters denotes literal output from GDB. Currently there is no need for GDB to output a newline followed by two ‘`control-z`’ characters, but if there was such a need, the annotations could be extended with an ‘`escape`’ annotation which means those three characters as output.

A simple example of starting up GDB with annotations is:

```
$ gdb --annotate=2
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and you are welcome to change it and/or distribute copies of it
under certain conditions.
Type "show copying" to see the conditions.
There is absolutely no warranty for GDB. Type "show warranty"
for details.
This GDB was configured as "sparc-sun-sunos4.1.3"

^Z^Zpre-prompt
(gdb)
^Z^Zprompt
quit

^Z^Zpost-prompt
$
```

Here ‘`quit`’ is input to GDB; the rest is output from GDB. The three lines beginning ‘`^Z^Z`’ (where ‘`^Z`’ denotes a ‘`control-z`’ character) are annotations; the rest is output from GDB.

2 Limitations of the Annotation Interface

The level two annotations mechanism is known to have a number of technical and architectural limitations. As a consequence, in 2001, with the release of GDB 5.1 and the addition of GDB/MI, the annotation interface was marked as deprecated.

This chapter discusses the known problems.

2.1 Dependant on CLI output

The annotation interface works by interspersing markups with GDB normal command-line interpreter output. Unfortunately, this makes the annotation client dependant on not just the annotations, but also the CLI output. This is because the client is forced to assume that specific GDB commands provide specific information. Any change to GDB’s CLI output modifies or removes that information and, consequently, likely breaks the client.

Since the GDB/MI output is independent of the CLI, it does not have this problem.

2.2 Scalability

The annotation interface relies on value annotations (see Chapter 5 [Value Annotations], page 3) and the display mechanism as a way of obtaining up-to-date value information. These mechanisms are not scalable.

In a graphical environment, where many values can be displayed simultaneously, a serious performance problem occurs when the client tries to first extract from GDB, and then re-display, all those values. The client should instead only request and update the values that changed.

The GDB/MI Variable Objects provide just that mechanism.

2.3 Correctness

The annotation interface assumes that a variable's value can only be changed when the target is running. This assumption is not correct. A single assignment to a single variable can result in the entire target, and all displayed values, needing an update.

The GDB/MI Variable Objects include a mechanism for efficiently reporting such changes.

2.4 Reliability

The GDB/MI interface includes a dedicated test directory (`gdb/gdb.mi`), and any addition or fix to GDB/MI must include testsuite changes.

2.5 Maintainability

The annotation mechanism was implemented by interspersing CLI print statements with various annotations. As a consequence, any CLI output change can alter the annotation output.

Since the GDB/MI output is independent of the CLI, and the GDB/MI is increasingly implemented independent of the CLI code, its long term maintenance is much easier.

3 Migrating to GDB/MI

By using the `'interp mi'` command, it is possible for annotation clients to invoke GDB/MI commands, and hence access the GDB/MI. By doing this, existing annotation clients have a migration path from this obsolete interface to GDB/MI.

4 The Server Prefix

To issue a command to GDB without affecting certain aspects of the state which is seen by users, prefix it with `'server '`. This means that this command will not affect the command history, nor will it affect GDB's notion of which command to repeat if `RET` is pressed on a line by itself.

The server prefix does not affect the recording of values into the value history; to print a value without recording it into the value history, use the `output` command instead of the `print` command.

5 Values

Value Annotations have been removed. GDB/MI instead provides Variable Objects.

When a value is printed in various contexts, GDB uses annotations to delimit the value from the surrounding text.

If a value is printed using `print` and added to the value history, the annotation looks like

```
^Z^Zvalue-history-begin history-number value-flags
history-string
^Z^Zvalue-history-value
the-value
^Z^Zvalue-history-end
```

where *history-number* is the number it is getting in the value history, *history-string* is a string, such as '\$5 = ', which introduces the value to the user, *the-value* is the output corresponding to the value itself, and *value-flags* is '*' for a value which can be dereferenced and '-' for a value which cannot.

If the value is not added to the value history (it is an invalid float or it is printed with the `output` command), the annotation is similar:

```
^Z^Zvalue-begin value-flags
the-value
^Z^Zvalue-end
```

When GDB prints an argument to a function (for example, in the output from the `backtrace` command), it annotates it as follows:

```
^Z^Zarg-begin
argument-name
^Z^Zarg-name-end
separator-string
^Z^Zarg-value value-flags
the-value
^Z^Zarg-end
```

where *argument-name* is the name of the argument, *separator-string* is text which separates the name from the value for the user's benefit (such as '='), and *value-flags* and *the-value* have the same meanings as in a `value-history-begin` annotation.

When printing a structure, GDB annotates it as follows:

```
^Z^Zfield-begin value-flags
field-name
^Z^Zfield-name-end
separator-string
^Z^Zfield-value
the-value
^Z^Zfield-end
```

where *field-name* is the name of the field, *separator-string* is text which separates the name from the value for the user's benefit (such as '='), and *value-flags* and *the-value* have the same meanings as in a `value-history-begin` annotation.

When printing an array, GDB annotates it as follows:

```
^Z^Zarray-section-begin array-index value-flags
```

where *array-index* is the index of the first element being annotated and *value-flags* has the same meaning as in a *value-history-begin* annotation. This is followed by any number of elements, where is element can be either a single element:

```
' , ' whitespace      ; omitted for the first element
the-value
^Z^Zelt
```

or a repeated element

```
' , ' whitespace      ; omitted for the first element
the-value
^Z^Zelt-rep number-of-repetitions
repetition-string
^Z^Zelt-rep-end
```

In both cases, *the-value* is the output for the value of the element and *whitespace* can contain spaces, tabs, and newlines. In the repeated case, *number-of-repetitions* is the number of consecutive array elements which contain that value, and *repetition-string* is a string which is designed to convey to the user that repetition is being depicted.

Once all the array elements have been output, the array annotation is ended with

```
^Z^Zarray-section-end
```

6 Frames

Value Annotations have been removed. GDB/MI instead provides a number of frame commands.

Frame annotations are no longer available. The GDB/MI provides ‘-stack-list-arguments’, ‘-stack-list-locals’, and ‘-stack-list-frames’ commands.

Whenever GDB prints a frame, it annotates it. For example, this applies to frames printed when GDB stops, output from commands such as `backtrace` or `up`, etc.

The frame annotation begins with

```
^Z^Zframe-begin level address
level-string
```

where *level* is the number of the frame (0 is the innermost frame, and other frames have positive numbers), *address* is the address of the code executing in that frame, and *level-string* is a string designed to convey the level to the user. *address* is in the form ‘0x’ followed by one or more lowercase hex digits (note that this does not depend on the language). The frame ends with

```
^Z^Zframe-end
```

Between these annotations is the main body of the frame, which can consist of

-

```
^Z^Zfunction-call
function-call-string
```

where *function-call-string* is text designed to convey to the user that this frame is associated with a function call made by GDB to a function in the program being debugged.

-

```
^Z^Zsignal-handler-caller
signal-handler-caller-string
```

where *signal-handler-caller-string* is text designed to convey to the user that this frame is associated with whatever mechanism is used by this operating system to call a signal handler (it is the frame which calls the signal handler, not the frame for the signal handler itself).

- A normal frame.

This can optionally (depending on whether this is thought of as interesting information for the user to see) begin with

```
^Z^Zframe-address
address
^Z^Zframe-address-end
separator-string
```

where *address* is the address executing in the frame (the same address as in the **frame-begin** annotation, but printed in a form which is intended for user consumption—in particular, the syntax varies depending on the language), and *separator-string* is a string intended to separate this address from what follows for the user's benefit.

Then comes

```
^Z^Zframe-function-name
function-name
^Z^Zframe-args
arguments
```

where *function-name* is the name of the function executing in the frame, or ‘??’ if not known, and *arguments* are the arguments to the frame, with parentheses around them (each argument is annotated individually as well, see Chapter 5 [Value Annotations], page 3).

If source information is available, a reference to it is then printed:

```
^Z^Zframe-source-begin
source-intro-string
^Z^Zframe-source-file
filename
^Z^Zframe-source-file-end
:
^Z^Zframe-source-line
line-number
^Z^Zframe-source-end
```

where *source-intro-string* separates for the user's benefit the reference from the text which precedes it, *filename* is the name of the source file, and *line-number* is the line number within that file (the first line is line 1).

If GDB prints some information about where the frame is from (which library, which load segment, etc.; currently only done on the RS/6000), it is annotated with

```
^Z^Zframe-where
information
```

Then, if source is to actually be displayed for this frame (for example, this is not true for output from the **backtrace** command), then a **source** annotation (see Chapter 13 [Source Annotations], page 9) is displayed. Unlike most annotations, this is output instead of the normal text which would be output, not in addition.

7 Displays

Display Annotations have been removed. GDB/MI instead provides Variable Objects.

When GDB is told to display something using the `display` command, the results of the display are annotated:

```
^Z^Zdisplay-begin
number
^Z^Zdisplay-number-end
number-separator
^Z^Zdisplay-format
format
^Z^Zdisplay-expression
expression
^Z^Zdisplay-expression-end
expression-separator
^Z^Zdisplay-value
value
^Z^Zdisplay-end
```

where *number* is the number of the display, *number-separator* is intended to separate the number from what follows for the user, *format* includes information such as the size, format, or other information about how the value is being displayed, *expression* is the expression being displayed, *expression-separator* is intended to separate the expression from the text that follows for the user, and *value* is the actual value being displayed.

8 Annotation for GDB Input

When GDB prompts for input, it annotates this fact so it is possible to know when to send output, when the output from a given command is over, etc.

Different kinds of input each have a different *input type*. Each input type has three annotations: a **pre-** annotation, which denotes the beginning of any prompt which is being output, a plain annotation, which denotes the end of the prompt, and then a **post-** annotation which denotes the end of any echo which may (or may not) be associated with the input. For example, the **prompt** input type features the following annotations:

```
^Z^Zpre-prompt
^Z^Zprompt
^Z^Zpost-prompt
```

The input types are

- prompt** When GDB is prompting for a command (the main GDB prompt).
- commands** When GDB prompts for a set of commands, like in the `commands` command.
The annotations are repeated for each command which is input.
- overload-choice**
When GDB wants the user to select between various overloaded functions.
- query** When GDB wants the user to confirm a potentially dangerous operation.
- prompt-for-continue**
When GDB is asking the user to press return to continue. Note: Don't expect this to work well; instead use `set height 0` to disable prompting. This is because the counting of lines is buggy in the presence of annotations.

9 Errors

`^Z^Zquit`

This annotation occurs right before GDB responds to an interrupt.

`^Z^Zerror`

This annotation occurs right before GDB responds to an error.

Quit and error annotations indicate that any annotations which GDB was in the middle of may end abruptly. For example, if a `value-history-begin` annotation is followed by a `error`, one cannot expect to receive the matching `value-history-end`. One cannot expect not to receive it either, however; an error annotation does not necessarily mean that GDB is immediately returning all the way to the top level.

A quit or error annotation may be preceded by

`^Z^Zerror-begin`

Any output between that and the quit or error annotation is the error message.

Warning messages are not yet annotated.

10 Information on Breakpoints

Breakpoint Annotations have been removed. GDB/MI instead provides breakpoint commands.

The output from the `info breakpoints` command is annotated as follows:

```
^Z^Zbreakpoints-headers
header-entry
^Z^Zbreakpoints-table
```

where *header-entry* has the same syntax as an entry (see below) but instead of containing data, it contains strings which are intended to convey the meaning of each field to the user. This is followed by any number of entries. If a field does not apply for this entry, it is omitted. Fields may contain trailing whitespace. Each entry consists of:

```
^Z^Zrecord
^Z^Zfield 0
number
^Z^Zfield 1
type
^Z^Zfield 2
disposition
^Z^Zfield 3
enable
^Z^Zfield 4
address
^Z^Zfield 5
what
^Z^Zfield 6
frame
^Z^Zfield 7
condition
^Z^Zfield 8
ignore-count
^Z^Zfield 9
commands
```

Note that *address* is intended for user consumption—the syntax varies depending on the language.

The output ends with

```
^Z^Zbreakpoints-table-end
```

11 Invalidation Notices

The following annotations say that certain pieces of state may have changed.

`^Z^Zframes-invalid`

The frames (for example, output from the `backtrace` command) may have changed.

`^Z^Zbreakpoints-invalid`

The breakpoints may have changed. For example, the user just added or deleted a breakpoint.

12 Running the Program

When the program starts executing due to a GDB command such as `step` or `continue`,

```
^Z^Zstarting
```

is output. When the program stops,

```
^Z^Zstopped
```

is output. Before the `stopped` annotation, a variety of annotations describe how the program stopped.

`^Z^Zexited exit-status`

The program exited, and *exit-status* is the exit status (zero for successful exit, otherwise nonzero).

`^Z^Zsignalled`

The program exited with a signal. After the `^Z^Zsignalled`, the annotation continues:

```
intro-text
^Z^Zsignal-name
name
^Z^Zsignal-name-end
middle-text
^Z^Zsignal-string
string
^Z^Zsignal-string-end
end-text
```

where *name* is the name of the signal, such as `SIGILL` or `SIGSEGV`, and *string* is the explanation of the signal, such as `Illegal Instruction` or `Segmentation fault`. *intro-text*, *middle-text*, and *end-text* are for the user's benefit and have no particular format.

`^Z^Zsignal`

The syntax of this annotation is just like `signalled`, but GDB is just saying that the program received the signal, not that it was terminated with it.

`^Z^Zbreakpoint number`

The program hit breakpoint number *number*.

`^Z^Zwatchpoint number`

The program hit watchpoint number *number*.

13 Displaying Source

The following annotation is used instead of displaying source code:

`^Z^Zsource filename:line:character:middle:addr`

where *filename* is an absolute file name indicating which source file, *line* is the line number within that file (where 1 is the first line in the file), *character* is the character position within the file (where 0 is the first character in the file) (for most debug formats this will necessarily point to the beginning of a line), *middle* is ‘middle’ if *addr* is in the middle of the line, or ‘beg’ if *addr* is at the beginning of the line, and *addr* is the address in the target program associated with the source which is being displayed. *addr* is in the form ‘0x’ followed by one or more lowercase hex digits (note that this does not depend on the language).

14 Multi-threaded Applications

The following annotations report thread related changes of state.

`^Z^Znew-thread`

This annotation is issued once for each thread that is created apart from the main thread, which is not reported.

`^Z^Zthread-changed`

The selected thread has changed. This may occur at the request of the user with the `thread` command, or as a result of execution, e.g., another thread hits a breakpoint.

`^Z^Zthread-exited,id="id",group-id="gid"`

This annotation is issued once for each thread that exits. The *id* field contains the global GDB identifier of the thread. The *gid* field identifies the thread group this thread belongs to.

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Version 1.3, 3 November 2008

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